STPM 2021 PHYSICS 960/2 (U2)

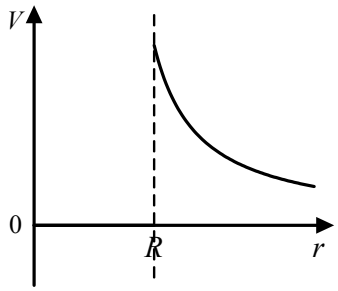
Questions and Answers

Values of constants: $e = 1.60 \times 10^{-19}$ C, $\mu_0 = 4\pi \times 10^{-7}$ H m^{-1}. Q20(b) is interpreted as the loop being moved completely out of the field in 0.050 s.

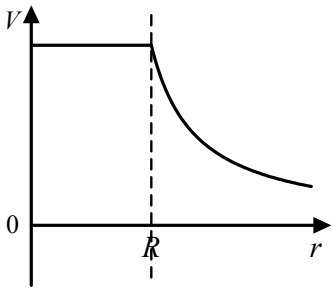
Section A [15 marks]

1. An isolated conducting sphere of radius R carries charge $-Q$. Which graph shows the variation of the electric potential, V, with distance, r, from the centre of the sphere?

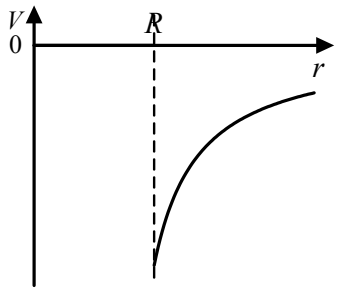
A



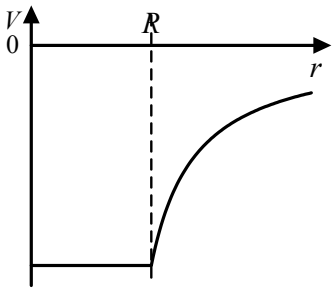
B



C



D

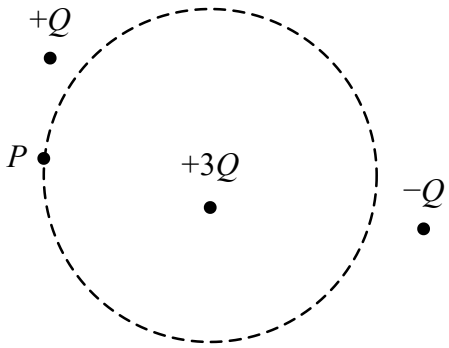


Answer: D

A negatively charged conductor has a constant negative potential everywhere inside (no field inside); outside, $V = -kQ/r$ rises towards zero (D).

Explanation: Potential is continuous at the surface; only the field jumps.

2. A system of charges, $+3Q$, $+Q$ and $-Q$, is distributed in a region as shown in the diagram below.



If a Gaussian surface is assumed to enclose the $+3Q$ and P is a point on the surface as shown in the above diagram, which is true about the system of the charges?

A The electric field strength at P depends on the $+3Q$ only.
B The electric field strength on the Gaussian surface is the same.
C The electric field strength inside the Gaussian surface is the same.
D The net flux through the Gaussian surface depends on the $+3Q$ only.

Answer: D

Gauss's law: net flux $= \frac{Q_{\text{enclosed}}}{\varepsilon_0} = \frac{3Q}{\varepsilon_0}$: it depends on the $+3Q$ only (D)

Explanation: The field at P depends on all three charges, and it is not uniform over the surface.

3. The capacitance of a parallel plate capacitor increases when

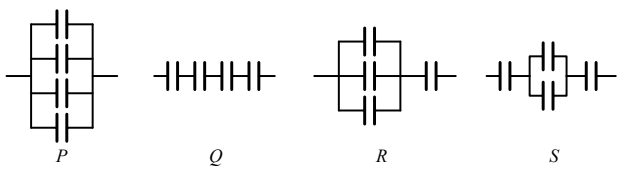
A smaller metal plates are used
B the separation of the plates is increased
C the potential difference across the plates is decreased
D an insulator is inserted into the space between the plates

Answer: D

$C = \frac{\varepsilon_r \varepsilon_0 A}{d}$: inserting an insulator (dielectric) increases ε_r and so C (D)

Explanation: C does not depend on the p.d.

4. Four identical capacitors are arranged in four different ways, P, Q, R and S, as shown in the diagram below.



Which arrangement shows the decreasing order of effective capacitance?

A P, R, S, Q B P, S, R, Q C Q, R, S, P D Q, S, R, P

Answer: A

P: four in parallel = $4C$. R: $3C$ in series with $C = 0.75C$. S: $2C$ in series with two single $C = 0.4C$.
Q: four in series = $0.25C$.
Decreasing order: P, R, S, Q.

Explanation: In series the smallest capacitance dominates.

5. When the potential difference between two points on a conductor is directly proportional to the distance, d, between the points, the electric field in the conductor is

A zero
B constant
C directly proportional to d^2
D inversely proportional to d^2

Answer: B

$$E = \frac{V}{d};\ V \propto d \Longrightarrow E \text{ is constant}$$

Explanation: A uniform potential gradient means a uniform field.

6. Which statement is true about the decrease in the electrical conductivity of a metal when its temperature increases?

A The density of free electrons decreases.
B The random speed of free electrons increases.
C The mean free path of free electrons decreases.
D The mean time between collision of free electrons and ions increases.

Answer: C

Greater lattice vibration means more frequent collisions, so the mean free path (and mean free time) of the electrons decreases (C).

Explanation: The number of free electrons in a metal is almost unchanged.

7. A battery of e.m.f. 12.0 V and an internal resistance 5.00×10^{-2} Ω is connected to a resistor of resistance 2.00 Ω. What is the potential difference and the power dissipated of the battery?

	Potential difference	*Power dissipated*
A	11.7 V	1.71 W
B	11.7 V	68.5 W
C	12.0 V	1.71 W
D	12.0 V	70.2 W

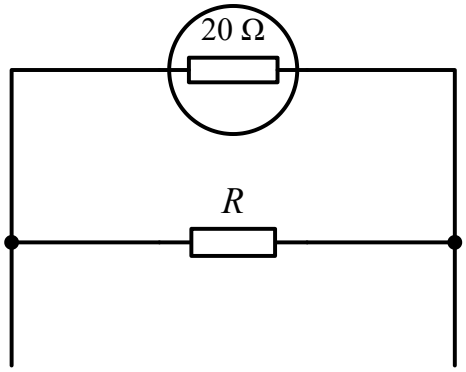
Answer: A

$$I = \frac{12.0}{2.00 + 0.0500} = 5.85 \text{ A; terminal p.d. } = IR = (5.85)(2.00) = 11.7 \text{ V}$$

Power dissipated in the battery $= I^2r = (5.85)^2(0.0500) = 1.71$ W

Explanation: The power delivered to the resistor is 68.5 W; the total is 70.2 W.

8. A shunt, R, which converts miliammeter to ammeter, is connected parallel to a miliammeter with resistance 20 Ω as shown in the diagram below.



The full-scale deflection of the miliammeter is 1 mA. If the required full-scale deflection of the ammeter is 1.0 A, what is the resistance of R?

A 0.002 Ω B 0.02 Ω C 20 Ω D 20 kΩ

Answer: B

Shunt current = $1.0 - 0.001 = 0.999$ A; $R = \frac{(0.001)(20)}{0.999} = 0.02$ Ω

Explanation: The shunt and the meter have the same p.d.

9. The magnitude of magnetic flux density is the magnetic force acting on a

A conductor which carries a current of 1 A perpendicularly with the magnetic field
B conductor which carries a current of 1 A along the direction of the magnetic field
C unit charge which moves at speed 1 m s^{-1} along the direction of the magnetic field
D unit charge which moves at speed 1 m s^{-1} perpendicularly with the magnetic field

Answer: A

$B = \frac{F}{IL \sin\theta}$: B equals the force on a conductor carrying 1 A perpendicular to the field (per unit length)

Explanation: The force on a unit charge moving at 1 m s^{-1} perpendicular to the field (D) is an equivalent definition and gives the same value; the intended answer is A. No force acts on a current or charge moving along the field.

10. A magnetic field is produced by a solenoid which carries a current of 10 A. If the solenoid has a length of 50 cm with 5000 turns, what is the magnetic field in the solenoid?

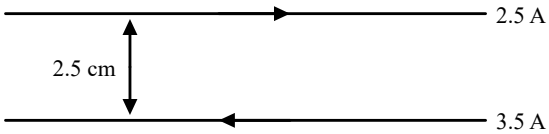
A 1.0×10^{-3} T B 6.0×10^{-3} T C 6.3×10^{-2} T D 1.3×10^{-1} T

Answer: D

$$B = \mu_0 \frac{N}{l} I = (4\pi \times 10^{-7}) \frac{5000}{0.50} (10) = 1.3 \times 10^{-1} \text{ T}$$

Explanation: $n = N/l$ is the number of turns per metre.

11. Two long parallel wires are 2.5 cm apart, carry currents of 2.5 A and 3.5 A respectively are shown in the diagram below.



Which is the force per unit length between both wires?

A Attractive with magnitude of 7.0×10^{-5} N m^{-1}
B Repulsive with magnitude of 7.0×10^{-5} N m^{-1}
C Attractive with magnitude of 2.2×10^{-4} N m^{-1}
D Repulsive with magnitude of 2.2×10^{-4} N m^{-1}

Answer: B

$$\frac{F}{l} = \frac{\mu_0 I_1 I_2}{2\pi d} = \frac{(4\pi \times 10^{-7})(2.5)(3.5)}{2\pi(0.025)} = 7.0 \times 10^{-5} \text{ N m}^{-1}$$

The currents are in opposite directions, so the force is repulsive.

Explanation: Like currents attract, unlike currents repel.

12. A bar magnet moves with a constant speed, v, into a coil and induces an e.m.f., E_1, in the coil. The bar magnet is then withdrawn via the same path with a constant speed of $\frac{1}{2}v$, and induces an e.m.f., E_2, in the coil. What is the relation between E_2 and E_1?

A $E_2 = -2E_1$ B $E_2 = 2E_1$ C $E_2 = -0.5E_1$ D $E_2 = 0.5E_1$

Answer: C

Withdrawing reverses the sign of the e.m.f.; half the speed halves the rate of change of flux:

$$E_2 = -0.5E_1$$

Explanation: $\varepsilon \propto \mathrm{d}\Phi/\mathrm{d}t$.

13. Two insulated wires are wound onto a common cardboard cylinder of length 0.50 m and cross-sectional area 0.050 m^2 to form two coils. The number of turns in the coils are 1500 and 500 respectively. What is the mutual inductance between the two coils?

A 31 mH B 94 mH C 280 mH D 1900 mH

Answer: B

$$M = \frac{\mu_0 N_1 N_2 A}{l} = \frac{(4\pi \times 10^{-7})(1500)(500)(0.050)}{0.50} = 9.4 \times 10^{-2}\ \text{H} = 94\ \text{mH}$$

Explanation: Both coils share the same cross-section, so all the flux links both.

14. A resistor of resistance R is connected to an alternating voltage supply. If the frequency of the voltage supply is doubled, what is the resistance of the resistor?

A $\frac{1}{4}R$ B $\frac{1}{2}R$ C R D $2R$

Answer: C

The resistance of a resistor does not depend on frequency: it stays R.

Explanation: Only reactances (X_L, X_C) depend on frequency.

15. An alternating current dynamo produces an electric power 200 kW at root mean square voltage 10.0 kV. The power is transmitted through a cable of resistance 5.00 Ω. What is the power dissipated by the cable?

A 200 kW B 198 kW C 2.00 kW D 0.500 kW

Answer: C

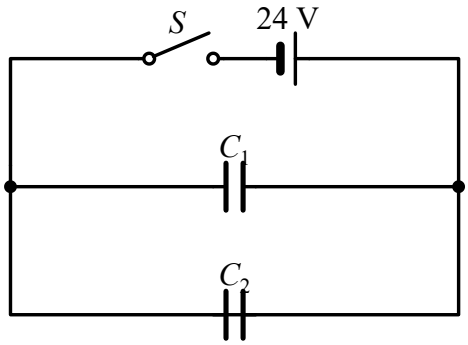
$$I_{\text{rms}} = \frac{P}{V_{\text{rms}}} = \frac{200 \times 10^3}{10.0 \times 10^3} = 20.0\ \text{A}$$

$$P_{\text{loss}} = I^2 R = (20.0)^2(5.00) = 2.00\ \text{kW}$$

Explanation: High-voltage transmission keeps the current, and so the loss, small.

Section B [15 marks]

16. Two air-filled parallel plate capacitors, C_1 and C_2, of capacitance 10 μF and 20 μF respectively are connected to a battery of e.m.f. 24 V and a switch, S, as shown in the circuit diagram below.



An insulator of dielectric constant 5.0 is then inserted into the space between the plates of C_1.

(a) Calculate the total capacitance of the circuit. [3]

Answer

$C_1' = 5.0 \times 10\ \mu F = 50\ \mu F$ 1

Parallel: $C = 50 + 20$ 1

$C = 70\ \mu F$ 1

3

Explanation: The dielectric multiplies only C_1.

(b) Determine the total charge stored in the capacitors. [2]

Answer

$Q = CV = (70 \times 10^{-6})(24)$ 1

$Q = 1.68 \times 10^{-3}$ C 1

2

Explanation: Both capacitors have the full 24 V across them.

(c) S is then opened and an uncharged capacitor of capacitance 5.0 μF is connected in parallel to C_2. Calculate the potential difference across the capacitors. [3]

Answer

With S open the total charge is conserved: 1.68×10^{-3} C 1

New total capacitance = 70 + 5.0 = 75 μF 1

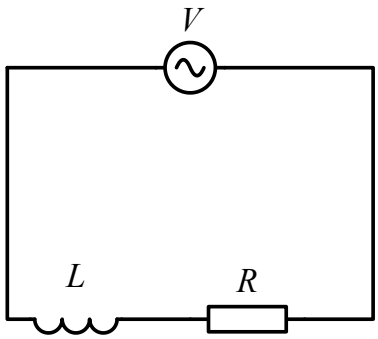
$V = \dfrac{1.68 \times 10^{-3}}{75 \times 10^{-6}} = 22.4$ V 1

3

Total for question **8**

Explanation: Charge flows into the new capacitor until all three share the same p.d.

17. A resistor, R, and an inductor, L, are connected in series to an alternating voltage supply, V, as shown in the circuit diagram below.



The voltage supply has peak voltage and oscillating frequency of 25 V and 50.0 Hz respectively. The resistance of the resistor is 15.0 Ω and the reactance of the inductor is 30.0 Ω.

(a) Determine the self inductance of the inductor. [2]

Answer

$L = \dfrac{X_L}{2\pi f} = \dfrac{30.0}{2\pi(50.0)}$ 1

$L = 0.0955$ H 1

2

Explanation: $X_L = 2\pi fL$.

(b) Calculate the impedance of the circuit. [2]

Answer

$Z = \sqrt{(R^2 + X_L{}^2)} = \sqrt{(15.0^2 + 30.0^2)}$ 1

$Z = 33.5\ \Omega$ 1

2

Explanation: Resistance and reactance add as phasors.

(c) Determine the root mean square current of the circuit. [3]

Answer

$V_{\text{rms}} = \dfrac{25}{\sqrt{2}} = 17.7$ V 1

$I_{\text{rms}} = \dfrac{V_{\text{rms}}}{Z} = \dfrac{17.7}{33.5}$ 1

$I_{\text{rms}} = 0.527$ A 1

3

Total for question **7**

Explanation: Convert the peak voltage to r.m.s. first.

Section C [30 marks]

18. (a) State Kirchhoff's laws. [2]

Answer

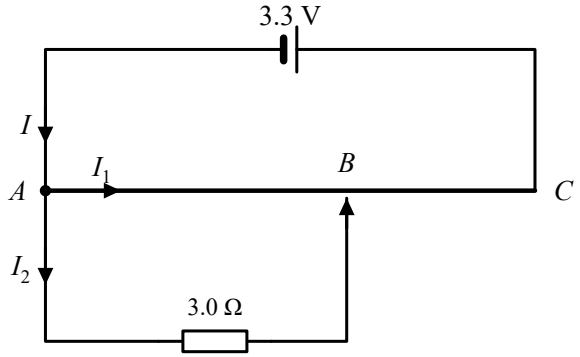
First law: the sum of currents entering a junction equals the sum of currents leaving it (conservation of charge) — 1

Second law: around any closed loop, the sum of e.m.f.s equals the sum of the p.d.s (IR) (conservation of energy) — 1

2

Explanation: $\Sigma I = 0$ at a junction; $\Sigma \varepsilon = \Sigma IR$ round a loop.

(b) A potential divider, which consists of 100 cm length uniform slides wire with resistance of 2.0 Ω, is used to supply variable voltage to a table lamp of resistance 3.0 Ω as shown in the circuit diagram below.



The voltage supply has an e.m.f. 3.3 V with negligible internal resistance.

(i) Determine the range of voltage supplied to the table lamp. [1]

Answer

0 V (contact at A) to 3.3 V (contact at C) — 1

1

Explanation: A potential divider can give any p.d. from zero up to the supply e.m.f.

(ii) If the length of the slide wire from A to B is 60.0 cm, determine the currents, I, I_1 and I_2. [8]

Answer

$R_{AB} = \frac{60.0}{100} \times 2.0 = 1.2\ \Omega$; $R_{BC} = 0.8\ \Omega$ 1

Lamp ‖ AB: $\frac{(1.2)(3.0)}{1.2 + 3.0}$ 1

$= 0.857\ \Omega$ 1

Total resistance $= 0.857 + 0.8 = 1.657\ \Omega$ 1

$I = \frac{3.3}{1.657} = 1.99$ A 1

P.d. across AB (and the lamp) $= (1.99)(0.857) = 1.71$ V 1

$I_1 = \frac{1.71}{1.2} = 1.42$ A 1

$I_2 = \frac{1.71}{3.0} = 0.569$ A 1

8

Explanation: Check: $I_1 + I_2 = I$.

(iii) Determine the power supplied to the table lamp. [2]

Answer

$P = I_2^2 R = (0.569)^2(3.0)$ 1

$P = 0.971$ W 1

2

Explanation: Equivalently $V^2/R = 1.71^2/3.0$.

(iv) Explain the change in the power supplied to the table lamp when the slide wire contact is moved towards C. [2]

Answer

Moving the contact towards C increases the p.d. across the lamp 1

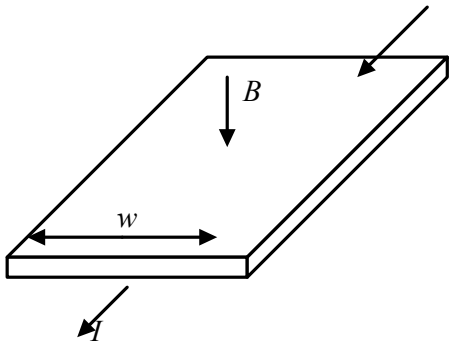
so the power supplied to the lamp increases (it becomes brighter) 1

2

Total for question **15**

Explanation: At C the lamp receives the full 3.3 V.

19. (a) A uniform magnetic field, B, is applied perpendicularly to the direction of current, I, in a semiconductor of width, w, as shown in the diagram below.



(i) Explain Hall effect of the semiconductor. [2]

Answer

The charge carriers moving in the magnetic field experience a magnetic force qvB sideways and collect on one face — 1

This builds up a p.d. (the Hall voltage) across the width, perpendicular to both I and B — 1

2

Explanation: The build-up stops when the electric force balances the magnetic force.

(ii) Derive an expression for the Hall voltage, V_H, across the semiconductor. [5]

Answer

Carriers deflect until the electric force balances the magnetic force: $qE = qvB$ — 1

$E = \frac{V_H}{w} \Rightarrow \frac{V_H}{w} = vB$ — 1

$I = nqvA$ with $A = wt$ (t = thickness) — 1

$v = \frac{I}{nqwt}$ — 1

$V_H = \frac{BI}{nqt}$ — 1

5

Explanation: V_H is larger when n is small, which is why semiconductors are used.

(iii) State **three** applications of the Hall effect in the semiconductor. [3]

Answer

Measuring magnetic flux density (Hall probe) — 1

Determining the type (sign) of charge carriers: n-type or p-type — 1

Measuring the number density of charge carriers — 1

3

Explanation: Hall sensors are also used as position and speed sensors.

(b) A thin slab *n*-type semiconductor of length, width and thickness are 12.5 mm, 7.50 mm and 2.50 mm respectively carries a current of 130 mA. A magnetic flux density 0.500 T produces a maximum Hall voltage of 9.15 mV between the edges of the slab.

(i) Calculate the number of free electrons per unit volume within the slab. [2]

Answer

$n = \dfrac{BI}{eV_H t} = \dfrac{(0.500)(130 \times 10^{-3})}{(1.60 \times 10^{-19})(9.15 \times 10^{-3})(2.50 \times 10^{-3})}$ 1

$n = 1.78 \times 10^{22}\ \text{m}^{-3}$ 1

2

Explanation: t is the thickness along **B** (2.50 mm).

(ii) Determine the magnetic force that acted on each electron. [3]

Answer

In equilibrium the magnetic force equals the electric force: $E = \dfrac{V_H}{w} = \dfrac{9.15 \times 10^{-3}}{7.50 \times 10^{-3}} = 1.22\ \text{V m}^{-1}$ 1

$F = eE = (1.60 \times 10^{-19})(1.22)$ 1

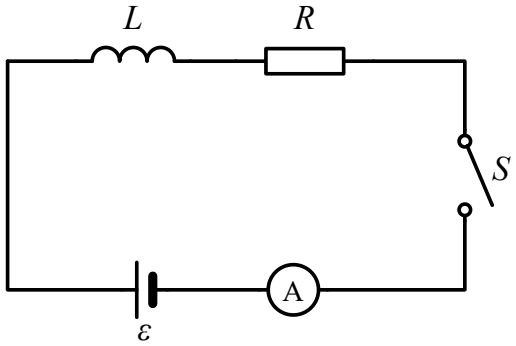
$F = 1.95 \times 10^{-19}\ \text{N}$ 1

3

Total for question **15**

Explanation: Equivalently $F = evB$ with $v = I/(nAe) = 2.44\ \text{m s}^{-1}$.

20. (a) A pure inductor, L, a resistor, R, a switch, S, and an ammeter, A, are connected to a battery, ε, of negligible internal resistance as shown in the circuit diagram below.



The self inductance, the resistance and the e.m.f. of the battery are 4.0 H, 1.6 Ω and 6.0 V respectively.

(i) When S is closed, the increment of the ammeter reading is slow. Explain the reasons to the slowness. [3]

Answer

When S is closed the current starts to increase, so the magnetic flux in the inductor increases — 1

A back e.m.f. is induced in the inductor (Faraday's law) — 1

By Lenz's law it opposes the growth of the current, so the current rises gradually (time constant $L/R = 2.5$ s) — 1

3

Explanation: The large inductance (4.0 H) makes the rise noticeably slow.

(ii) Determine the final current in the circuit. [2]

Answer

$I = \frac{\varepsilon}{R} = \frac{6.0}{1.6}$ — 1

$I = 3.75$ A — 1

2

Explanation: When the current is steady there is no back e.m.f.

(iii) Calculate the rate of change of current when the current reaches 2.0 A. [3]

Answer

$\varepsilon = IR + L\frac{dI}{dt}$ — 1

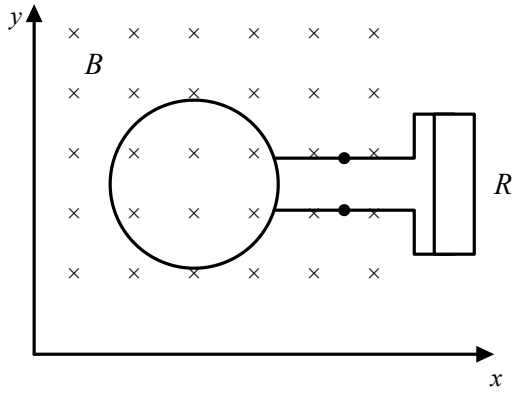
$\frac{dI}{dt} = \frac{6.0 - (2.0)(1.6)}{4.0}$ — 1

$= 0.70\ \text{A s}^{-1}$ — 1

3

Explanation: The rate of increase falls as the current grows.

(b) A circular loop wire which is connected to a resistor, R, is placed in a uniform magnetic field, B, directed into the plane of the page as shown in the diagram below.



The area of the loop is 33.5 cm^2, the magnitude of the magnetic fields is 1.35 T and the resistance of R is 1.0 Ω. The loop is moved up and down along the y-axis for 0.050 s.

(i) R becomes hot after the loop is moved. Explain your answer. [4]

Answer

As the loop moves out of the field, the magnetic flux through it changes	1
An e.m.f. is induced in the loop (Faraday's law)	1
The loop and R form a closed circuit, so an induced current flows	1
Electrical energy is dissipated as heat in R (I^2R), so R becomes hot	1
	4

Explanation: While the loop is completely inside a uniform field, moving it does not change the flux and no current flows.

(ii) Determine the current in the circuit. [3]

Answer

$\varepsilon = \frac{\Delta\Phi}{\Delta t} = \frac{BA}{\Delta t} = \frac{(1.35)(33.5 \times 10^{-4})}{0.050}$	1
$\varepsilon = 0.0905$ V	1
$I = \frac{\varepsilon}{R} = \frac{0.0905}{1.0} = 0.090$ A	1
	3
Total for question	**15**

Explanation: 33.5 cm^2 = 33.5 × 10^{-4} m^2.